Anchor Representation: 2024 vs. 2026

Overview

This report compares anchor representation between the 2024 and 2026 calibration datasets.

The purpose is to determine which anchors were sufficiently represented in each calibration period and which anchors were excluded because they did not meet the representation criterion.

For **both 2024 and 2026**, an anchor is considered sufficiently represented if it appears in at least **80% of the available calibration files**.

This analysis distinguishes between:

- anchors retained in both analyses;
- anchors retained in 2024 but not 2026;
- anchors retained in 2026 but not 2024;
- anchors that appeared in the raw data but were excluded from both analyses.

The analysis uses anchor IDs only. Calibration values are not required for this comparison.

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1. Load 2024 Calibration Files

The 2024 files are identified by their filename pattern.

```
[1] 29
```

The number above is the total number of available 2024 calibration files.

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2. Read 2024 Anchor IDs

Each 2024 file is read only to obtain the anchor IDs.

The IDs are immediately converted to character so that they have the same type used throughout the 2026 analysis.

Remove missing IDs:

---

3. Apply the 80% Representation Criterion to 2024

An anchor must appear in at least 80% of the available 2024 calibration files.

```
[1] 23.2
```

Count the number of files containing each anchor:

The anchors retained for the 2024 analysis are:

Number of retained 2024 anchors:

```
[1] 672
```

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4. Load 2026 Calibration Files

```
[1] 89
```

---

5. Remove 2026 Files with Fewer Than 500 Observations

This reproduces the filtering used in the primary 2026 calibration analysis.

```
[1] 80
```

The 2026 representation analysis therefore uses only files containing at least 500 observations, consistent with the primary 2026 analysis.

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6. Read 2026 Anchor IDs

Remove missing IDs:

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7. Apply the 80% Representation Criterion to 2026

An anchor must appear in at least 80% of the usable 2026 calibration files.

```
[1] 64
```

Count files containing each anchor:

The anchors retained for the 2026 analysis are:

Number of retained 2026 anchors:

```
[1] 899
```

---

8. Verify ID Types

Before comparing the datasets, explicitly force both ID columns to character.

Verify:

```
        dataset   id_type
1 retained_2024 character
2 retained_2026 character
3  raw_ids_2024 character
4  raw_ids_2026 character
```

All four should report `character`.

---

9. Anchors Retained in Both Years

These are anchors that meet the 80% criterion in both datasets.

Number retained in both:

```
[1] 535
```

View the IDs:

```
# A tibble: 535 x 1
   id
   <chr>
 1 6251
 2 6307
 3 6313
 4 6316
 5 6317
 6 6320
 7 6323
 8 6324
 9 6325
10 6327
# i 525 more rows
```

---

10. Anchors Retained in 2024 but Not 2026

These anchors met the 80% criterion in 2024 but did not meet it in 2026.

Number:

```
[1] 137
```

IDs:

```
# A tibble: 137 x 1
   id
   <chr>
 1 6025
 2 6039
 3 6043
 4 6054
 5 6072
 6 6082
 7 6087
 8 6088
 9 6092
10 6094
# i 127 more rows
```

---

11. Anchors Retained in 2026 but Not 2024

These anchors met the 80% criterion in 2026 but did not meet it in 2024.

Number:

```
[1] 364
```

IDs:

```
# A tibble: 364 x 1
   id
   <chr>
 1 6342
 2 6499
 3 6513
 4 6521
 5 6543
 6 6589
 7 6683
 8 6790
 9 6836
10 6868
# i 354 more rows
```

---

12. All Raw Anchors Observed in 2024

This represents every unique anchor that appeared at least once in the original 2024 calibration files.

Number:

```
[1] 691
```

---

13. All Raw Anchors Observed in 2026

This represents every unique anchor that appeared at least once in the usable 2026 calibration files.

Number:

```
[1] 965
```

---

14. Anchors Excluded from Both Analyses

An anchor is considered excluded from both analyses if:

- it appeared in the raw 2024 and/or 2026 data;
- it did not meet the 80% criterion in 2024;
- it did not meet the 80% criterion in 2026.

First create the complete set of anchors observed in either year.

Then remove every anchor retained in either analysis:

The resulting set consists of anchors excluded from both analyses:

Number excluded from both:

```
[1] 65
```

IDs:

```
# A tibble: 65 x 1
   id
   <chr>
 1 6042
 2 6120
 3 6295
 4 6430
 5 6508
 6 6657
 7 6738
 8 6886
 9 6939
10 6945
# i 55 more rows
```

---

15. Determine Where the Excluded Anchors Appeared

It is useful to distinguish anchors that were excluded from both but appeared in:

- 2024 only;
- 2026 only;
- both years.

15.1 Excluded from Both, Present in Both Raw Datasets

Number:

```
[1] 0
```

IDs:

```
# A tibble: 0 x 1
# i 1 variable: id <chr>
```

15.2 Excluded from Both, Present in 2024 Only

Number:

```
[1] 12
```

IDs:

```
# A tibble: 12 x 1
   id
   <chr>
 1 6042
 2 6120
 3 6295
 4 6430
 5 6508
 6 6886
 7 6939
 8 7089
 9 7194
10 7260
11 7266
12 7291
```

15.3 Excluded from Both, Present in 2026 Only

Number:

```
[1] 53
```

IDs:

```
# A tibble: 53 x 1
   id
   <chr>
 1 6657
 2 6738
 3 6945
 4 7332
 5 7362
 6 7377
 7 7411
 8 7444
 9 7490
10 7525
# i 43 more rows
```

---

16. Detailed Representation Table

The following table combines the representation information for every anchor observed in either year.

View the complete table:

```
# A tibble: 1,101 x 12
   id    files_2024 total_files_2024 proportion_2024 retained_2024 files_2026
   <chr>      <int>            <int>           <dbl> <lgl>              <int>
 1 6042          22               29           0.759 FALSE                 NA
 2 6120          22               29           0.759 FALSE                 NA
 3 6295          22               29           0.759 FALSE                 NA
 4 6430          14               29           0.483 FALSE                 NA
 5 6508          17               29           0.586 FALSE                 NA
```

```
 6 6886          20                29          0.690  FALSE                 NA
 7 6939          20                29          0.690  FALSE                 NA
 8 7089          17                29          0.586  FALSE                 NA
 9 7194           2                29          0.0690 FALSE                 NA
10 7260          23                29          0.793  FALSE                 NA
# i 1,091 more rows
# i 6 more variables: total_files_2026 <int>, proportion_2026 <dbl>,
#   retained_2026 <lgl>, present_2024 <lgl>, present_2026 <lgl>,
#   classification <chr>
```

---

17. Summary of Anchor Classification

```
# A tibble: 5 x 2
  classification                number_of_anchors
  <chr>                                     <int>
1 Retained in both                            535
2 Retained in 2026 only                       364
3 Retained in 2024 only                       137
4 Excluded from both; 2026 only                53
5 Excluded from both; 2024 only                12
```

---

18. Summary Table for the Report

A more compact summary is useful for reporting the results.

```
# A tibble: 6 x 2
  category                                         number_of_anchors
  <chr>                                                        <int>
1 Retained in both 2024 and 2026                                 535
2 Retained in 2024 only                                          137
3 Retained in 2026 only                                          364
4 Excluded from both; present in both raw datasets                 0
5 Excluded from both; present in 2024 only                        12
6 Excluded from both; present in 2026 only                        53
```

---

19. Representation Percentages

The following calculates the percentage of all anchors observed in either raw dataset that fall into each category.

```
# A tibble: 6 x 3
  category                                number_of_anchors percentage_of_all_ob~1
  <chr>                                               <int>                  <dbl>
1 Retained in both 2024 and 2026                        535                 0.486
2 Retained in 2024 only                                 137                 0.124
3 Retained in 2026 only                                 364                 0.331
4 Excluded from both; present in both ~                   0                 0
5 Excluded from both; present in 2024 ~                  12                 0.0109
6 Excluded from both; present in 2026 ~                  53                 0.0481
# i abbreviated name: 1: percentage_of_all_observed
```

---

20. Compare 80% Representation Rates

This table summarizes the number of anchors observed in each year and the number that passed the 80% criterion.

```
# A tibble: 2 x 6
  year  total_files eighty_percent_threshold raw_anchors retained_80pct
  <chr>       <int>                    <dbl>       <int>          <int>
1 2024           29                       24         691            672
2 2026           80                       64         965            899
# i 1 more variable: proportion_retained <dbl>
```

---

21. Anchors Retained in One Year but Not the Other

This table is particularly useful for identifying changes in anchor availability between calibration campaigns.

```
# A tibble: 501 x 2
   id    classification
   <chr> <chr>
 1 6025  Retained in 2024 only
 2 6039  Retained in 2024 only
 3 6043  Retained in 2024 only
 4 6054  Retained in 2024 only
 5 6072  Retained in 2024 only
 6 6082  Retained in 2024 only
 7 6087  Retained in 2024 only
 8 6088  Retained in 2024 only
 9 6092  Retained in 2024 only
10 6094  Retained in 2024 only
# i 491 more rows
```

---

22. Anchors Excluded from Both but Observed in Either Dataset

This is the group of anchors that may be particularly useful for investigating why anchors failed the representation criterion.

```
# A tibble: 65 x 8
   id    files_2024 total_files_2024 proportion_2024 files_2026 total_files_2026
   <chr>      <int>            <int>           <dbl>      <int>            <int>
 1 6042          22               29          0.759          NA               NA
 2 6120          22               29          0.759          NA               NA
 3 6295          22               29          0.759          NA               NA
 4 6430          14               29          0.483          NA               NA
 5 6508          17               29          0.586          NA               NA
 6 6886          20               29          0.690          NA               NA
 7 6939          20               29          0.690          NA               NA
 8 7089          17               29          0.586          NA               NA
 9 7194           2               29          0.0690         NA               NA
10 7260          23               29          0.793          NA               NA
# i 55 more rows
# i 2 more variables: proportion_2026 <dbl>, classification <chr>
```

---

23. Most Poorly Represented Anchors

The following table identifies anchors with the lowest representation in either year.

This can help identify anchors that were particularly intermittent.

```
# A tibble: 50 x 5
   id    representation_2024 representation_2026 minimum_representation
   <chr>               <dbl>               <dbl>                  <dbl>
 1 6025                1                       0                      0
 2 6039                1                       0                      0
 3 6042                0.759                   0                      0
 4 6043                1                       0                      0
 5 6054                1                       0                      0
 6 6072                1                       0                      0
 7 6082                1                       0                      0
 8 6087                1                       0                      0
 9 6088                1                       0                      0
10 6092                1                       0                      0
# i 40 more rows
# i 1 more variable: classification <chr>
```

---

24. Export Results

The major results are written to CSV files so they can be examined outside of the PDF report.

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25. Final Interpretation

The 80% representation criterion is applied independently to the two calibration periods.

An anchor retained in 2024 therefore appeared in at least 80% of the available 2024 calibration files, while an anchor retained in 2026 appeared in at least 80% of the usable 2026 calibration files.

The comparison separates three scientifically distinct groups:

1. **Stable/repeated anchors:** retained in both 2024 and 2026.

2. **Year-specific anchors:** retained in one calibration period but not the other.
3. **Intermittent anchors:** observed in the raw data but excluded because they failed the 80% representation criterion.

The final group can be particularly informative because failure to meet the 80% criterion may itself reflect changes in anchor availability, connectivity, geographic coverage, or other properties of the calibration system.

```
# A tibble: 6 x 2
  category                                             number_of_anchors
  <chr>                                                            <int>
1 Retained in both 2024 and 2026                                     535
2 Retained in 2024 only                                              137
3 Retained in 2026 only                                              364
4 Excluded from both; present in both raw datasets                     0
5 Excluded from both; present in 2024 only                            12
6 Excluded from both; present in 2026 only                            53
```

```
### One important detail

This version applies the criterion symmetrically, but **the denominator is not necessarily th

So, for example, if there are 100 usable 2024 files, an anchor needs at least 80 appearances

That is the appropriate interpretation of "present in at least 80% of the available files" an

The resulting **`representation_comparison`** table is probably the most valuable object proc
```